Applied Physics Letters ARTICLE pubs.aip.org/aip/apl

Reduction of magnetoelastic constant B_2 in Fe thin films by nitrogen implantation, as determined by surface acoustic waves

Cite as: Appl. Phys. Lett. **128**, 062404 (2026); doi: 10.1063/5.0289237
Submitted: 6 July 2025 · Accepted: 13 January 2026 ·
Published Online: 10 February 2026

View Online Export Citation CrossMark

A. Sharma, L. Christienne, F. Millo, M. Eddrief, E. Dandeu, J.-Y. Duquesne, C. Gourdon, L. Thevenard, P. Rovillain,[a] and M. Marangolo

AFFILIATIONS

Sorbonne Université, CNRS, Institut des Nanosciences de Paris, INSP, F-75005 Paris, France

Note: This paper is part of the Special Topic, Strain-controlled Magnetism: From Fundamental Phenomena to Devices.

[a]**Author to whom correspondence should be addressed:** pauline.rovillain@insp.upmc.fr

ABSTRACT

This article presents an original approach to both modify and measure the magnetoelastic coupling constant B_2 in epitaxial nitrogen-implanted Fe thin films presenting the $\alpha'-Fe_8N_{1-x}$ phase. Once the magnetocrystalline anisotropy has been determined by standard techniques, we measure the field dependence of Rayleigh surface acoustic wave (SAW) velocity variations. The magnetoelastic coupling constant is extracted by fitting the field-angle dependence of these variations. Using the magnetic and magnetoelastic parameters thus determined, the variations in SAW velocity as a function of the external field seen in surface acoustic wave induced ferromagnetic resonance experiments can be accurately reproduced, both in pristine and nitrogen-implanted Fe thin films. This ensemble of results confirms a reduction by almost half of the magnetoelastic coupling under doping level 2×10^{16} N_2/cm^2.

In recent years, particular attention has been given to the magnetoelastic coupling between surface acoustic waves (SAWs) and magnetic media.[1] This renewed interest echoes early observations made nearly fifty years ago,[2] which had already highlighted the potential of acoustic waves for magnetic sensing.[3] While initially underexplored, this line of research has experienced a revival, driven by new perspectives in spintronics and magnonics, as well as by promising applications in areas such as magnetic field sensing and acoustic wave-based devices.[4–6] In this context, recent research has focused on the surface acoustic wave-assisted ferromagnetic resonance (SAW-FMR) effect, where SAWs efficiently couple to magnetization dynamics.[7–13] SAW-FMR coupling becomes particularly significant when the external magnetic field B_{ext} satisfies the resonance condition, i.e., $B_{ext} = B_{res}$, leading to an intersection between dispersion relations of the SAWs and spin waves (SWs). This resonance is characterized by a sharp change in SAW velocity and increased acoustic absorption, resulting from enhanced SW excitation.[14] This well-understood phenomenon arises from the dynamic effective magnetic field induced by the Rayleigh wave via both magnetoelasticity (ME) and magneto-rotation (MR).[12] Magnetoelastic coupling is characterized by the magnetoelastic constants that reflect the coupling between magnetization and strain. Modifying these constants allows the ME coupling to be optimized for applications such as stress sensors, piezomagnetic actuators, or magnonics devices.[5,6,15,16] This coupling can be tuned during growth via alloying (NiFe, GaMnAs[17]) or atomic substitution (Fe–Ga).[18] The measurement of this ME coupling constant requires the use of advanced approaches, as detecting a shift in the resonant field is difficult, particularly for thin films. Indeed, the most reliable and widely used technique remains the optical or the capacitive cantilever method, which detects the curvature of the substrate and the clamped thin film under an applied magnetic field.[19–22] However, this technique presents significant experimental constraints. In particular, for thin films, this requires the substrate to be sufficiently thin to allow bending elastically with a measurable curvature change.[19]

In this article, we present an alternative way to tune magnetoelasticity after material growth in a controlled manner, i.e., by N-implantation. To obtain the magnetoelastic constant B_2, we develop a method relying on field-dependent SAW propagation in a magnetic layer. This approach allows us to accurately investigate the dependence of the SAW velocity on the magnetic field, especially in the SAW-FMR regime. We apply this method to a system that has been extensively studied in the literature: iron with interstitial nitrogen atoms (Fe:N).

In particular, we investigate epitaxial layers grown on a GaAs(001) substrates in the presence of the $\alpha' - Fe_8N_{1-x}$ ($x \sim 0$) phase. This system was chosen for two main reasons. First, the insertion of nitrogen preserves the epitaxy of iron on the substrate.[23,24] It thus offers an ideal playground, as no radical changes occur in the crystal structure apart from a tetragonal distortion. The subtle variations in magnetoelastic coupling can be systematically studied as a function of nitrogen insertion into the pristine bcc structure of iron. Second, FeN alloys are attracting considerable interest in the scientific community, due to the still debated increase in the magnetic moment of the ordered phase, i.e., the so-called $\alpha'' - Fe_8N$—a property that could, one day, enable the replacement of rare-earth elements in specific applications.[25] In addition, this study allows us to track how nitrogen implantation subtly modifies the SAW-FMR condition, thereby opening possibilities for magnonic SAW-based devices in which resonances can be locally tuned via controlled nitrogen insertion as recently proposed in Ref. 26.

In Fig. 1, we show a sketch of the sample under study. A single-crystal Fe thin film of thickness $t = 54$ nm was deposited through Molecular Beam Epitaxy (MBE) on a ZnSe (001) layer of 10 nm, grown on a GaAs (001) substrate. The epitaxial conditions are then Fe(001)//ZnSe(001)//GaAs(001) and Fe[100]//ZnSe[100]//GaAs[100]. The ZnSe layer prevents diffusion of Ga atoms into the Fe layer.[27] An Au capping layer of 12 nm was deposited to prevent oxidation of the Fe layer.[28] Two pristine Fe samples were implanted with different concentrations of N_2^+ at the IN2P3 JANNuS platform, using a 190 kV Nier–Bernas ion implanter. The acceleration energy was set to 26 keV, with selected doses of 1.0 and $2.0 \times 10^{16}\ N_2/cm^2$. Simulations using the Transport of Ions in Matter (TRIM) software indicate that the nitrogen implantation is not uniform across the Fe film; N ions are predominantly concentrated in the upper region of the Fe layer, just beneath the Au/Fe interface.[29,30] The pristine Fe sample and the implanted ones are subsequently etched into a mesa of a few millimeter square for interdigitated transducers (IDTs) lithography on the piezoelectric GaAs substrate.[31] More details on the sample can be found in Refs. 14 and 10. Table I lists the magnetic parameters, i.e., the in-plane cubic anisotropy (K_c) and out-of-plane uniaxial anisotropy (K_u), of the three samples. These values are obtained by Vector Network Analyzer-based Ferromagnetic Resonance (VNA-FMR) and out-of-plane Vibrating Sample Magnetometry (VSM) measurements, which are included in the supplementary material. As discussed in previous articles,[23,24,32] the increase in K_u with nitrogen implantation is accompanied by a decrease in K_c. X-ray diffraction (XRD) measurements presented in

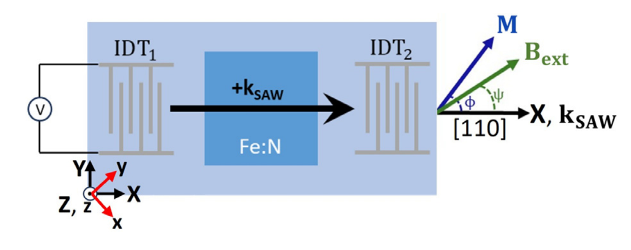

FIG. 1. Sample under study. The SAWs are generated using Al Split-52 IDTs at $f_{SAW} = 1.8$ GHz by applying an input signal at IDT$_1$. $\mathbf{B_{ext}}$ is the in-plane applied magnetic field, and $\mathbf{M}$ is the Fe magnetization. x // [100], y // [010], and z // [001] (red axes) correspond to the cubic frame, and x and y are the magnetic easy axes. X // [110], Y // [−110], and Z // [001] (black axes) correspond to the rotated frame with X and Y being the magnetic hard axes.

TABLE I. In-plane cubic (K_c) and out-of-plane uniaxial (K_u) anisotropy constants. K_c is deduced from VNA-FMR experiments. K_u is calculated by fitting the out-of-plane VSM measurements using the Stoner–Wolfarth model.

N_2 dose level	K_c (kJ/m^3)	K_u (kJ/m^3)
Pristine Fe	50 ± 2	50 ± 50
Fe:N ($1.0 \times 10^{16}\ N_2/cm^2$)	43 ± 3	200 ± 70
Fe:N ($2.0 \times 10^{16}\ N_2/cm^2$)	34 ± 7	355 ± 80

the supplementary material reveal that the former is due to nitrogen insertion into the octahedral sites of Fe[33] leading to tetragonal distorted bcc phase (α^*–Fe:N) and to the $\alpha' - Fe_8N_{1-x}$ phase. In contrast, we attribute the reduction in K_c to a slight increase in crystal mosaicity induced by nitrogen implantation. Notably, XRD measurements in the supplementary material indicate that the mosaicity remains low (of the order of 1°) and the film preserves its epitaxial conditions after implantation.

SAWs are generated and detected with split-52 IDTs,[34] at an acoustic frequency of $f_{SAW} = 1.8$ GHz. Pulsed excitation of 200 ns duration every 200 μs is used, and the acoustic signal is acquired with a digital oscilloscope to make direct time domain measurements. In the following, we show the relative SAW velocity variation, $\frac{\Delta V}{V}$, obtained with SAWs excited along the [110] direction ($\mathbf{k_{SAW}}//[110]$) as a function of an external magnetic field, B_{ext}, for various angles, ψ between $\mathbf{B_{ext}}$ and $\mathbf{k_{SAW}}$. $\frac{\Delta V}{V}$ is defined as $\frac{V(B_{ext}) - V(B_{ref})}{V(B_{ref})} = \frac{\beta(B_{ext}) - \beta(B_{ref})}{2\pi f_{SAW} \tau_{FeN}}$.[35] Here, β is the phase of the acoustic signal detected at IDT$_2$, and τ_{FeN} is the SAW travel time in the Fe:(N) layer. The reference field, B_{ref}, was set equal to 320 mT for all measurements, ensuring a complete magnetic saturation of magnetization along the applied magnetic field. Starting from B_{ref}, the magnetic field was ramped *downward* to prevent hysteretic SAW-FMR effects.[36] We study $\frac{\Delta V}{V}$ for different levels of N-implantation doses and interpret the evolution of the SAW-SW interaction as a function of the N dose.

It has been established that SAW-SW interaction is governed by magnetoelasticity (ME) and magneto-rotation (MR).[7,12,37] Their energy density terms normalized to the saturation magnetization (M_s) can be written as follows:

$$U_{cub}^{ME} = \frac{B_1}{M_s}\left(\varepsilon_{xx}\, m_x^2 + \varepsilon_{yy}\, m_y^2 + \varepsilon_{zz}\, m_z^2\right) + \frac{2B_2}{M_s}\left(\varepsilon_{xy}\, m_x\, m_y + \varepsilon_{yz}\, m_y\, m_z + \varepsilon_{xz}\, m_x\, m_z\right), \quad (1)$$

$$U_{Fe}^{MR} = \zeta \mu_0 M_s m_z \left(\omega_{yz}\, m_y + \omega_{xz}\, m_x\right), \quad (2)$$

where ε_{ii} refers to the strain in the *i* direction, ε_{ij} is the shear strain associated with directions i and j, $m_i = M_i/M_s$ denotes the reduced magnetization components along the *i* direction where $i = x, y, z$ and ω_{ij} are the anti-symmetric elements of the strain tensor representing infinitesimal local rotations associated with directions i and j. Equation (2) should, in principle, include correction terms related to K_c and K_u.[38] However, since $\mu_0 M_s^2 \approx 3.6 \times 10^6\ J/m^3$ in Fe is much larger than K_u and K_c terms, these corrections can be safely neglected in our case.

The effective magnetic field acting on the magnetization can be derived from the energy density U from the following equation: $\mathbf{b} = -\nabla_{\mathbf{m}} U$. In the rotated frame for SAWs propagating along the

[110] direction, the effective magnetic field corresponding to the SAW-SW interaction (b_{ME-MR}) is given by

$$b_\phi = -\frac{B_2 \varepsilon_{XX}}{M_s} \sin(2\phi_0), \tag{3}$$

$$b_\theta = -\left(\frac{2\, B_2\, \varepsilon_{XZ}}{M_s} + \zeta \mu_0 M_s\, \omega_{XZ}\right) \cos(\phi_0), \tag{4}$$

where b_ϕ and b_θ are the in-plane and out-of-plane components of b_{ME-MR} and *X*, *Y*, and *Z* axes are parallel to [110], [−110], and [001] directions, respectively. It is important to note that the magnetoelastic effective field is exclusively determined by the magnetoelastic coupling constant B_2, which quantifies the strength of the interaction under the deformation induced by SAWs propagating along the [110] direction. We discuss in the supplementary material how a similar experiment with the SAW propagating along [100] would allow for the determination of the longitudinal magnetoelastic coupling constant, B_1. The factor ζ in b_θ is equal to 0.45[39] and has been taken from Ref. 40. Physically, ζ is a measure of the dipolar field generated over the two faces of the thin film, averaged over its thickness.

Since the relative SAW velocity variation is of the order of 10^{-5}, the SAW dispersion is only slightly perturbed, and the coupling regime is weak. An important consequence of the weak coupling regime is that the acoustic power transmitted to the magnetic layer (ΔP) and the relative SAW velocity variation are related directly as follows:[14,37]

$$\frac{\Delta V}{V} = \frac{R_{eff}}{2\, k_{SAW}\, d_{Fe}\, P_{ac}^{layer}} \mathrm{Re}(\Delta P), \tag{5}$$

where

$$\Delta P = -\frac{\omega M_s}{2\mu_0} \int_{V_0} (b_\theta^*, b_\phi^*)\, \overleftrightarrow{\chi} \begin{pmatrix} b_\theta \\ b_\phi \end{pmatrix} dV. \tag{6}$$

In Eq. (5), P_{ac}^{layer} is the total power carried by the SAW in the magnetic layer, and d_{Fe} is the propagation distance of the SAW in the magnetic layer. In Eq. (6), V_0 and $\overleftrightarrow{\chi}$ denote the volume of the Fe layer and the Polder matrix, respectively. The dimensionless factor R_{eff} in Eq. (5) is a scaling factor for the available magnetic power in the layer and is directly proportional to the SAW frequency. Its value is determined for pristine Fe[14] and then kept indentical for the implanted samples. The complete phenomenological approach has been discussed in Ref. 14.

Figures 2(a) and 2(b) show the experimental relative variation in SAW velocities for the three samples at $f_{SAW} = 1.8$ GHz and their calculated SW frequencies for $k_{SW} = 3.9$ rad/μm as a function of the external field, B_{ext}, at $\psi = 0°$, respectively. The data exhibit three distinct magnetic field regions:

(i) High-field region: The magnetization vector is aligned along $B_{ext} \parallel$ [110]. The SW frequencies lie well above the SAW frequency, and a weak variation of SAW velocity is observed.

(ii) SAW-FMR region: The magnetization gradually rotates toward the in-plane easy axes (i.e., the [100] and/or [010] directions), leading to a softening of the SW frequency. This softening causes the SW frequency to intersect the SAW frequency at two distinct field values, resulting in a sharp $\frac{\Delta V}{V}$ drop characterized by two dips. As previously reported in Ref. 14, although not shown in this article, the sharp transition and appearance of the dips occur only when $B_{ext} \parallel k_{SAW} \parallel$ [110]. A slight misalignment of just 0.2° leads to a broadening of the transition and disappearance of the two dips, as the SW frequencies increase and the resonance conditions are no longer fulfilled. In contrast to our previous work in Ref. 14, these two dips are observed thanks to an increase in SAW frequency to 1.8 GHz.

(iii) Low-field region: The magnetization fully aligns along the easy axes, leading to an increase in SW frequencies and an overall increase of the SAW velocity as the resonance conditions are no longer satisfied.

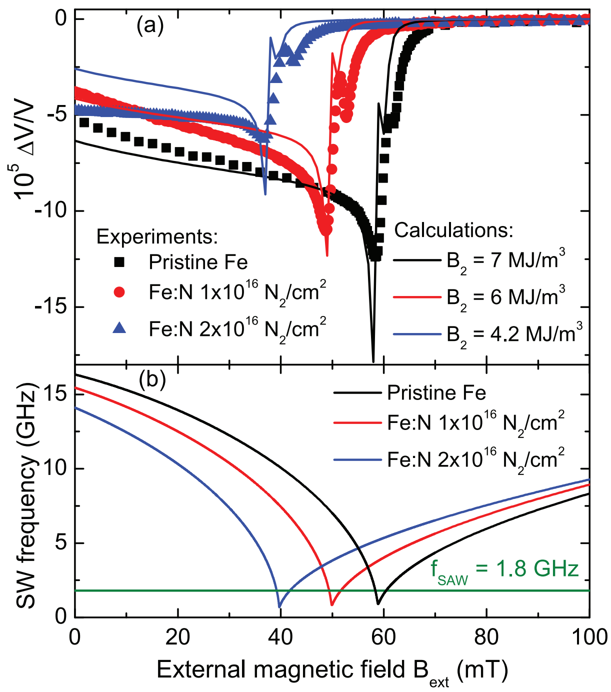

FIG. 2. (a) Experimental (symbols) and calculated (curves) SAW relative velocity variation vs in-plane external magnetic field, for pristine Fe (black square), intermediate (red circle) and highest (blue triangle) implanted doses, Fe:N, for $\psi = 0°$ at $f_{SAW} = 1.8$ GHz. (b) Calculated spin wave frequency for $k_{SW} = 3.9$ rad/μm vs in-plane external magnetic field. The green line indicates the SAW frequency at 1.8 GHz.

The first key experimental result of this study is that nitrogen implantation reduces the in-plane magnetic anisotropy K_c (see Table I), thereby shifting the SAW-SW resonance field toward lower values as reflected in Fig. 2(b). As a consequence, the SAW-SW resonance condition experiences a rigid shift: the sharp variation in relative velocity, marked by the two characteristic dips, moves toward lower magnetic field values with increasing doping, as shown in Fig. 2(a).

We then propose a method to evaluate the magnetoelastic constant B_2 by measuring the SAW velocity as a function of the angle ψ between the SAW propagation direction and the applied magnetic field. Specifically, we record $\frac{\Delta V}{V}$ as a function of B_{ext} for angles ψ ranging from 0° to 360° in steps of 10°, and then plot the experimental $\frac{\Delta V}{V}$ as a function of ψ at a fixed external field (see Fig. 3).

We select $B_{ext} = 150$ mT, corresponding to complete magnetic saturation, since a close examination of Eqs. (3)–(6) shows that, under

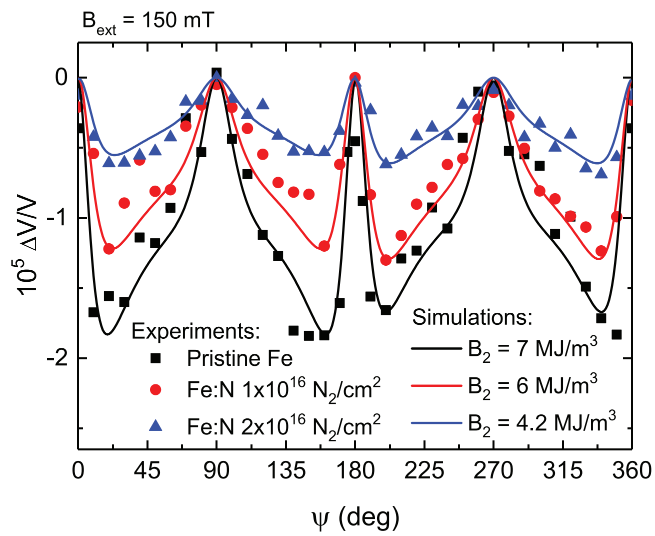

FIG. 3. SAW relative velocity variation ($\frac{\Delta V}{V}$) vs ψ for pristine Fe (black, square), intermediate (red, circle) and highest (blue, triangle) doped Fe:N at $B_{ext} = 150$ mT. At this field, the magnetization is completely saturated along the field direction. The solid line curves denote the calculated $\frac{\Delta V}{V}$ vs ψ for different values of B_2 as a function of the N dose and at a fixed $R_{eff} = 0.22$.

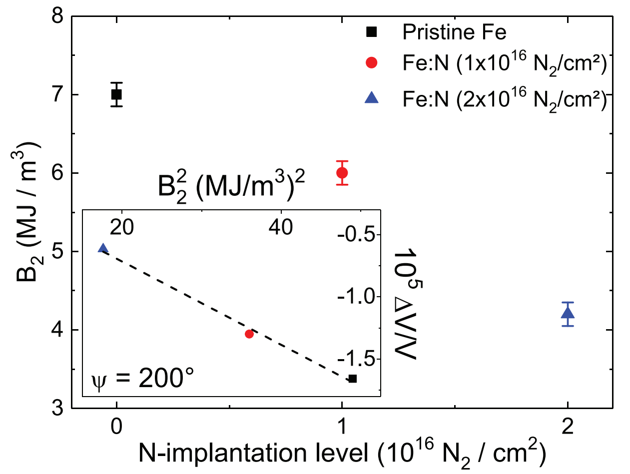

FIG. 4. Evolution of the magnetoelastic constant B_2 with N-implantation dose in Fe. Inset: linear dependence of $\Delta V/V$ with B_2^2 at the saturation field of 150 mT, taken at $\psi = 200°$ for the three samples. The main source of error in B_2 is due to the error in M_s, which stems from uncertainty in the magnetic material thickness as explained in Ref. 29.

saturation, $\frac{\Delta V}{V}$ depends only on the magnetoelastic coefficient B_2. In Fig. 3, we simulate the experimental data of pristine Fe by fixing $B_2 = 7$ MJ/m^3 (value already known[41,42]) and by adjusting the parameter R_{eff} to obtain good agreement between the experimental points and the theoretical curve.[43] It is worth noting that the term ζ, introduced in the driving fields of Eq. (4), effectively captures the asymmetry observed in the $\frac{\Delta V}{V}$ vs ψ measurements as shown in the supplementary material. In particular, the asymmetry in velocity variation values on either side of the main peak at $\psi = 180°$ is accurately reproduced.

For the two implanted samples, we fix $R_{eff} = 0.22$ determined by fitting the measurements on pristine Fe, and adjust B_2 to fit their $\frac{\Delta V}{V}$ vs ψ measurements. The experimental results and the corresponding calculations reveal that B_2 decreases with increasing doping concentration, as illustrated in Fig. 3. The B_2 values are 7, 6, and 4.2 MJ/m^3 for implantation levels of 0 (pristine sample), 10^{16}, and 2×10^{16} N_2/cm^2, respectively, and are presented in Fig. 4. The corresponding magnetostriction coefficients can easily be computed using $\lambda_{111} = -B_2/3C_{44}$, assuming the nitrogen implantation does not modify the elastic constant of Fe ($C_{44} = 115.9$ GPa): $|\lambda_{111}| = 20.1$, 17.2, and 12 ppm for doping levels of 0, 10^{16}, and 2×10^{16} N_2/cm^2, respectively. Note that values of up to $\lambda_{100} \approx -120$ ppm have been recorded for the Fe_4N phase though.[44,45]

In the following, we corroborate the magnetic parameters extracted from experiments by comparing the measured curves with calculated ones [Fig. 2(a)] using Eq. (5). The simulations are based on the magnetic anisotropy constants listed in Table I and the B_2 values reported in Fig. 4. These results demonstrate that the experimentally extracted parameters form a consistent and reliable set for reproducing the field dependence of $\frac{\Delta V}{V}$, as shown in Fig. 2(a), in particular the position of the resonance. This supports the validity of our approach, which consists in measuring B_2 at high magnetic fields—i.e., outside the SAW-FMR resonance regime—in order to decouple the measurements from the influence of magnetic anisotropy constants.

However, in Fig. 2(a), a deviation is observed between the calculated and experimental $\frac{\Delta V}{V}$ vs external field curves in the low field region, where the measured SAW velocity variation is significantly lower than predicted. This discrepancy stems from the presence of perpendicular magnetic anisotropy (PMA), which is not fully accounted for in our model. The effect is particularly pronounced in the highest dose sample, where PMA can lead to the formation of weak magnetic stripe domains.[24] Additionally, some discrepancies are observed near the resonance field, which may arise from factors not included in the model, such as the mosaicity of the epitaxied thin film (see the supplementary material) and a possible slight misalignment of the applied magnetic field known to be critical for SAW-FMR on Fe epitaxied on GaAs(001).[10]

Moreover, the inset of Fig. 4 shows a linear dependence between $\frac{\Delta V}{V}$ and B_2^2 in agreement with the high field expression of $\frac{\Delta V}{V}$, the procedure used to obtain this plot is detailed in the supplementary material.

In the following, we put forward a few hypotheses to explain the observed reduction in the magnetoelastic coefficient B_2 with increasing nitrogen content. Let us recall that nitrogen implantation in epitaxial Fe stabilizes the so-called $\alpha' - Fe_8N_{1-x}$ phase. Nitrogen atoms occupy octahedral interstitial sites, leading to an increase in Fe–Fe nearest-neighbor distances.[46] In our Fe:N thin films, epitaxy on GaAs(001) is preserved, with a slight increase in the in-plane lattice parameter (2.87 Å compared to 2.86 Å for pristine Fe) and a pronounced out-of-plane tetragonal distortion ($c = 3.11$ and 3.14 Å for intermediate and highest N-implantation doses, respectively), as previously reported.[24,29]

We then address the microscopic origin of magnetoelastic coupling. To second order in the spin density field and first order in the displacement field associated with long-wavelength acoustic phonons, the magnetoelastic energy and coefficients B_i can be written as the sum of two terms:[47] a first term that arises from relativistic effects, such as dipole–dipole interactions and spin–orbit coupling, and a second term that becomes non-zero only when the magnetization is nonuniform. This second term vanishes for a uniformly magnetized Fe lattice but can become non-zero when a nitrogen atom is introduced into the unit cell, perturbing the collinearity of the spins. In the case of the $\alpha'-Fe_8N_{1-x}$ phase, *ab initio* calculations have shown that the orbital magnetic moment of Fe is strongly influenced by the presence of N atoms, leading to the emergence of perpendicular magnetic anisotropy (PMA), as experimentally observed in our sample.[33] We thus hypothesize that the observed reduction in the B_2 coefficient originates from the same mechanism as PMA: a variation in the orbital moment of Fe atoms and a modification of the crystal field and of relativistic terms caused by the presence of N atoms in the unit cell. This remains a tentative interpretation and should be considered with caution. Detailed calculations based on the tetragonally distorted $\alpha'-Fe_8N_{1-x}$ phase would be required to confirm this mechanism.

In conclusion, this work demonstrates the possibility of modifying the SAW-FMR conditions and magnetoelastic coupling by implanting nitrogen after Fe growth. This was demonstrated by determining the magnetoelastic constant B_2 by an alternative method using SAW-FMR experiments. The magnetoelastic constant B_2 is extracted by investigating the angular dependency of SAW velocity variation $\Delta V/V$ at saturating magnetic field. Beyond this main result, we have also validated two theoretical predictions: first, the role of the dipolar fields generated on the two surfaces of the thin film in producing the asymmetry observed in SAW propagation; second, that $\Delta V/V$ varies linearly with B_2^2, the squared magnetoelastic constant. These insights contribute to a deeper understanding of magnetoelastic coupling in nitrogen-implanted Fe systems and provide a robust framework to study future SAW-based technological applications.

See the supplementary material for details on the characterization of samples and the extraction of anisotropy parameters, $\Delta V/V$ dependence with B_2^2, the influence of magneto-rotation on the SAW-SW driving fields, and the experimental geometry used to determine B_1, along with a table of acoustic parameters.

P.R., F.M., and M.M. acknowledge funding from the French National Research Agency, ANR-22-CE24-0015 SACOUMAD. The authors also acknowledge support from the France 2030 program PEPR SPIN, under Contract Nos. ANR-22-EXSP-0008 and ANR-22-EXSP-0004. The authors thank Paola Atkinson for preparing the GaAs substrate by MBE; Franck Fortuna for N-implantation at the IN2P3 JANNuS platform; Serge Vincent, Yunlin Zheng, Jean-Eudes Duvauchelle, and the staff of the MPBT (physical properties—low temperature platform of Sorbonne Université) for their support; and Mathieu Desquirez for developing the VNA-FMR post-processing script.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

A. Sharma: Formal analysis (equal); Writing – original draft (equal); Writing – review & editing (equal). **L. Christienne:** Formal analysis (supporting). **F. Millo:** Formal analysis (equal); Writing – original draft (equal); Writing – review & editing (equal). **M. Eddrief:** Investigation (equal); Resources (equal). **E. Dandeu:** Resources (equal). **J.-Y. Duquesne:** Conceptualization (equal); Formal analysis (equal); Funding acquisition (equal); Methodology (equal). **C. Gourdon:** Methodology (equal); Writing – review & editing (equal). **L. Thevenard:** Formal analysis (equal); Funding acquisition (equal); Methodology (equal); Project administration (equal); Supervision (equal); Writing – review & editing (equal). **P. Rovillain:** Conceptualization (equal); Formal analysis (equal); Funding acquisition (equal); Methodology (equal); Project administration (equal); Supervision (equal); Writing – original draft (equal); Writing – review & editing (equal). **M. Marangolo:** Conceptualization (equal); Funding acquisition (equal); Methodology (equal); Supervision (equal); Writing – original draft (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

REFERENCES

[1] J. Puebla, Y. Hwang, S. Maekawa, and Y. Otani, "Perspectives on spintronics with surface acoustic waves," Appl. Phys. Lett. **120**, 220502 (2022).
[2] A. Ganguly, K. Davis, D. Webb, C. Vittoria, and D. Forester, "Magnetically tuned surface-acoustic-wave phase shifter," Electron. Lett. **11**, 610–611 (1975).
[3] A. Ganguly, K. Davis, D. Webb, and C. Vittoria, "Magnetoelastic surface waves in a magnetic film–piezoelectric substrate configuration," J. Appl. Phys. **47**, 2696–2704 (1976).
[4] P. Smole, W. Ruile, C. Korden, A. Ludwig, E. Quandt, S. Krassnitzer, and P. Pongratz, "Magnetically tunable saw-resonator," in *Proceedings of the IEEE International Frequency Control Symposium and PDA Exhibition Jointly with the 17th European Frequency and Time Forum* (IEEE, 2003), pp. 903–906.
[5] A. Barman, G. Gubbiotti, S. Ladak, A. O. Adeyeye, M. Krawczyk, J. Gräfe, C. Adelmann, S. Cotofana, A. Naeemi, V. I. Vasyuchka *et al.*, "The 2021 magnonics roadmap," J. Phys.: Condens. Matter **33**, 413001 (2021).
[6] P. Delsing, A. N. Cleland, M. J. A. Schuetz, J. Knörzer, G. Giedke, J. I. Cirac, K. Srinivasan, M. Wu, K. C. Balram, C. Bäuerle, T. Meunier, C. J. B. Ford, P. V. Santos, E. Cerda-Méndez, H. Wang, H. J. Krenner, E. D. S. Nysten, M. Weiß, G. R. Nash, L. Thevenard, C. Gourdon, P. Rovillain, M. Marangolo, J.-Y. Duquesne, G. Fischerauer, W. Ruile, A. Reiner, B. Paschke, D. Denysenko, D. Volkmer, A. Wixforth, H. Bruus, M. Wiklund, J. Reboud, J. M. Cooper, Y. Fu, M. S. Brugger, F. Rehfeldt, and C. Westerhausen, "The 2019 surface acoustic waves roadmap," J. Phys. D **52**, 353001 (2019).
[7] M. Weiler, L. Dreher, C. Heeg, H. Huebl, R. Gross, M. S. Brandt, and S. T. B. Goennenwein, "Elastically driven ferromagnetic resonance in nickel thin films," Phys. Rev. Lett. **106**, 117601 (2011).
[8] L. Thevenard, C. Gourdon, J.-Y. Prieur, H. J. Von Bardeleben, S. Vincent, L. Becerra, L. Largeau, and J.-Y. Duquesne, "Surface-acoustic-wave-driven ferromagnetic resonance in (Ga, Mn)(As,P) epilayers," Phys. Rev. B **90**, 094401 (2014).
[9] A. Hernández-Mínguez, F. Macià, J. Hernàndez, J. Herfort, and P. Santos, "Large nonreciprocal propagation of surface acoustic waves in epitaxial ferromagnetic/semiconductor hybrid structures," Phys. Rev. Appl. **13**, 044018 (2020).
[10] J.-Y. Duquesne, P. Rovillain, C. Hepburn, M. Eddrief, P. Atkinson, A. Anane, R. Ranchal, and M. Marangolo, "Surface-acoustic-wave induced ferromagnetic resonance in Fe thin films and magnetic field sensing," Phys. Rev. Appl. **12**, 024042 (2019).
[11] P. G. Gowtham, T. Moriyama, D. C. Ralph, and R. A. Buhrman, "Traveling surface spin-wave resonance spectroscopy using surface acoustic waves," J. Appl. Phys. **118**, 233910 (2015).
[12] M. Xu, K. Yamamoto, J. Puebla, K. Baumgaertl, B. Rana, K. Miura, H. Takahashi, D. Grundler, S. Maekawa, and Y. Otani, "Nonreciprocal surface acoustic wave propagation via magneto-rotation coupling," Sci. Adv. **6**, eabb1724 (2020).

[13]F. Millo, R. L. Seeger, C. Chappert, A. Solignac, and T. Devolder, "Symmetry of the dissipation of surface acoustic waves by ferromagnetic resonance," AIP Adv. **15**, 045230 (2025).
[14]P. Rovillain, J.-Y. Duquesne, L. Christienne, M. Eddrief, M. G. Pini, A. Rettori, S. Tacchi, and M. Marangolo, "Impact of spin-wave dispersion on surface-acoustic-wave velocity," Phys. Rev. Appl. **18**, 064043 (2022).
[15]X. Liang, C. Dong, H. Chen, J. Wang, Y. Wei, M. Zaeimbashi, Y. He, A. Matyushov, C. Sun, and N. Sun, "A review of thin-film magnetoelastic materials for magnetoelectric applications," Sensors **20**, 1532 (2020).
[16]M. Matsuo, E. Saitoh, and S. Maekawa, "Spin-mechatronics," J. Phys. Soc. Jpn. **86**, 011011 (2017).
[17]S. C. Masmanidis, H. X. Tang, E. B. Myers, M. Li, K. De Greve, G. Vermeulen, W. Van Roy, and M. L. Roukes, "Nanomechanical measurement of magnetostriction and magnetic anisotropy in (Ga,Mn)As," Phys. Rev. Lett. **95**, 187206 (2005).
[18]E. M. Summers, T. A. Lograsso, and M. Wun-Fogle, "Magnetostriction of binary and ternary Fe–Ga alloys," J. Mater. Sci. **42**, 9582–9594 (2007).
[19]D. Sander, "The correlation between mechanical stress and magnetic anisotropy inultrathin films," Rep. Prog. Phys. **62**, 809 (1999).
[20]M. Coïsson, W. Hüttenes, M. Cialone, G. Barrera, F. Celegato, P. Rizzi, Z. H. Barber, and P. Tiberto, "Measurement of thin film magnetostriction using field-dependent atomic force microscopy," Appl. Surf. Sci. **525**, 146514 (2020).
[21]F. Zighem and D. Faurie, "A review on nanostructured thin films on flexible substrates: Links between strains and magnetic properties," J. Phys.: Condens. Matter **33**, 233002 (2021).
[22]J.-P. Jay, F. Le Berre, S. Pogossian, and M. Indenbom, "Direct and inverse measurement of thin films magnetostriction," J. Magn. Magn. Mater. **322**, 2203–2214 (2010).
[23]T. Amarouche, L.-C. Garnier, M. Marangolo, M. Eddrief, V. H. Etgens, F. Fortuna, Y. Sadaoui, M. Tamine, J. Cantin, and H. von Bardeleben, "Influence of ion implantation parameters on the perpendicular magnetic anisotropy of Fe-N thin films with stripe domains," J. Appl. Phys. **121**, 243903 (2017).
[24]L.-C. Garnier, M. Eddrief, S. Fin, D. Bisero, F. Fortuna, V. H. Etgens, and M. Marangolo, "Perpendicular magnetic anisotropy in Fe–N thin films: Threshold field for irreversible magnetic stripe domain rotation," Spin **6**, 1640014 (2016).
[25]J.-P. Wang, "Environment-friendly bulk $Fe_{16}N_2$ permanent magnet: Review and prospective," J. Magn. Magn. Mater. **497**, 165962 (2020).
[26]I. Ngouagnia Yemeli, L. Christienne, P. Rovillain, J.-Y. Duquesne, A. Anane, M. Marangolo, and D. Stoeffler, "A micromagnetic study of surface-acoustic-wave-driven excitation of spin waves in an iron-based conduit," J. Appl. Phys. **137**, 153908 (2025).
[27]M. Marangolo, F. Gustavsson, M. Eddrief, P. Sainctavit, V. H. Etgens, V. Cros, F. Petroff, J. George, P. Bencok, and N. Brookes, "Magnetism of the Fe/ZnSe (001) interface," Phys. Rev. Lett. **88**, 217202 (2002).
[28]I. S. Camara, S. Tacchi, L. C. Garnier, M. Eddrief, F. Fortuna, G. Carlotti, and M. Marangolo, "Magnetization dynamics of weak stripe domains in Fe–N thin films: A multi-technique complementary approach," J. Phys.: Condens. Matter **29**, 465803 (2017).
[29]L. Christienne, J. Jiménez-Bustamante, P. Rovillain, M. Eddrief, Y. Zheng, F. Fortuna, M. Marangolo, M. Madami, R. Gallardo, P. Landeros, and S. Tacchi, "Nonreciprocal spin-wave propagation in anisotropy-graded iron films prepared by nitrogen implantation," Phys. Rev. Appl. **24**, 064040 (2025).
[30]L.-C. Garnier, "Couches minces en Fe-N élaborées par implantation ionique: Propriétés structurales et magnétiques," Ph.D. thesis [Université Paris Saclay (COmUE), 2019].
[31]The size of the Fe mesa is $2 \times 0.5\,\mathrm{mm}^2$, and the delay line for this sample is $1000\,\mu\mathrm{m}$. For the N-implanted samples, the mesa are $2 \times 2\,\mathrm{mm}^2$ each, with delay lines of $2500\,\mu\mathrm{m}$. Split-52 IDTs were used in our experiment with the following parameters: acoustic aperture $(w) = 450\,\mu\mathrm{m}$, number of electrode pairs $(N_p) = 25$, fundamental frequency $(f_0) = 895\,\mathrm{MHz}$, SAW velocity in GaAs $(v_{SAW}) = 2767$ m/s.
[32]P. Gupta, H. Fiedler, S. Rubanov, and J. Kennedy, "Magnetisation and magnetic anisotropy of ion beam synthesised iron nitride," J. Magn. Magn. Mater. **517**, 167388 (2021).
[33]H. Zhang, I. Dirba, T. Helbig, L. Alff, and O. Gutfleisch, "Engineering perpendicular magnetic anisotropy in Fe via interstitial nitrogenation: N choose *K*," APL Mater. **4**, 116104 (2016).
[34]F. J. Schülein, E. Zallo, P. Atkinson, O. G. Schmidt, R. Trotta, A. Rastelli, A. Wixforth, and H. J. Krenner, "Fourier synthesis of radiofrequency nanomechanical pulses with different shapes," Nat. Nanotechnol. **10**, 512–516 (2015).
[35]Since the phase β is a function of the external field, sweeping the field downward from B_{ref} to 0 leads to a phase shift. At fixed frequency, this phase shift corresponds to a change in k_{SAW}, and therefore, to a change in the acoustic velocity. By taking the relative velocity variation $(\frac{\Delta V}{V})$, we subtract any contribution of the velocity variation arising from the substrate. Thus, the SAW travel time (τ_{FeN}) is taken only for the magnetic layer.
[36]A. K. Vythelingum, T. Tremblais, D. B. Nguyen, D. Ourdani, E. Dandeu, V. Laude, C. Gourdon, and L. Thevenard, "Role of coercivity in surface acoustic wave driven ferromagnetic resonance," ACS Appl. Electron. Mater. **7**, 4055–4064 (2025).
[37]L. Dreher, M. Weiler, M. Pernpeintner, H. Huebl, R. Gross, M. S. Brandt, and S. T. Goennenwein, "Surface acoustic wave driven ferromagnetic resonance in nickel thin films: Theory and experiment," Phys. Rev. B **86**, 134415 (2012).
[38]E. D. Trémolet de Lacheisserie, *Magnetostriction: Theory and Applications of Magnetoelasticity* (CRC Press, 1993).
[39]ζ is a dimensionless factor equal to $\frac{1-e^{-|k_{SAW}|t}}{2|k_{SAW}|t}$, and it quantifies the dipolar field produced by the two film surfaces that survives after averaging through the thickness t, and this factor should be accounted for to obtain the correct weight of the asymmetry observed in $\frac{\Delta V}{V}$ of the SAW propagation.
[40]K. Yamamoto and S. Maekawa, "Magnetostatic field induced by mechanical deformations," Ann. Phys. **536**, 2300395 (2024).
[41]M. Barturen, D. Sander, J. Milano, J. Premper, C. Helman, M. Eddrief, J. Kirschner, and M. Marangolo, "Bulklike behavior of magnetoelasticity in epitaxial $Fe_{1-x}Ga_x$ thin films," Phys. Rev. B **99**, 134432 (2019).
[42]G. Bayreuther, J. Premper, M. Sperl, and D. Sander, "Uniaxial magnetic anisotropy in Fe/GaAs (001): Role of magnetoelastic interactions," Phys. Rev. B **86**, 054418 (2012).
[43]We checked that R_{eff} does not vary notably with B_2 by using the fully magnetoelastic approach described at length in Ref. 14. We found that R_{eff} varies by less than 10^{-4} when B_2 varies from 7 to 1 $\mathrm{MJ/m^3}$.
[44]K. Sin and S. Wang, "Stress and magnetic properties in high moment FeN thin films," J. Appl. Phys. **79**, 5901–5903 (1996).
[45]K. Ito, I. Kurniawan, Y. Shimada, Y. Miura, Y. Endo, and T. Seki, "Giant tunability of magnetoelasticity in Fe_4N system as a platform to unveil correlation between magnetostriction and magnetic damping," Commun. Mater. **6**, 53 (2025).
[46]J. Coey and P. Smith, "Magnetic nitrides," J. Magn. Magn. Mater. **200**, 405–424 (1999).
[47]A. Rückriegel, P. Kopietz, D. A. Bozhko, A. A. Serga, and B. Hillebrands, "Magnetoelastic modes and lifetime of magnons in thin yttrium iron garnet films," Phys. Rev. B **89**, 184413 (2014).